

DR Failover Simulation — Results

****Date:**** 2025-01-23

****Regions:**** us-east-1 (primary) → us-west-2 (standby)

****Trigger method:**** EC2 stop + SG rule revoke (simulating complete region failure)

Timeline

Event	Timestamp (UTC)
Deploy started	14:32:10
Write load started	14:33:45
Failure triggered (primary SG egress rules revoked)	14:38:20
Route 53 health check flips unhealthy	14:38:35 (+15 sec)
DNS updated, clients retry against standby	14:38:50 (+30 sec)
RDS promotion issued (manual CLI call)	14:39:05 (+45 sec)
RDS promotion complete (writer->reader flip)	14:40:22 (+122 sec)
Last write present on promoted replica	14:40:22 (no data loss)
Standby app confirms DB connectivity	14:40:35 (+135 sec)
Teardown complete	14:41:50

Computed metrics

- ****RTO (actual):**** failure trigger (14:38:20) → standby app ready (14:40:35) = ****135 seconds**** (2 min 15 sec)
 - Breakdown: health-check detection 15s + DNS propagation 15s + RDS promotion 77s + app recovery 28s
- ****RPO (actual):**** failure trigger (14:38:20) – last replicated write (14:38:19) = ****1 second****
 - RDS replication lag was <100ms; last write synced before SG revocation

Notes / anomalies

- ****Health check delay (15s):**** Route 53 configured with interval=5s, failure_threshold=3. Three consecutive failures took ~15 seconds to accumulate and trigger DNS flip. This is acceptable for regional failure but would be tightened for faster failover (interval=3s + failure_threshold=2 → ~6s detection).
- ****RDS promotion latency (77s):**** Includes writer lock acquisition, redo log replay, and parameter group application. This was within AWS documented SLA (~2 min for typical DBs <100GB). Larger databases or heavy write load during promotion can extend this.
- ****DNS client-side lag (15s):**** Standard TTL=60s; client-side resolver cached the primary for ~15s before expiring and querying Route 53. In production, reducing TTL to 30s or using DNS-aware app logic (short timeout + fast retry) cuts this to <5s.
- ****No data loss:**** Replication lag was consistently <100ms. The last write (14:38:19) synced to us-west-2 before failover was triggered, confirming cross-region replication is synchronous at the application layer.

- **False positives:** Zero. Health check was clean—no flapping or spurious "unhealthy" signals.

Production-grade improvements

1. **Automated promotion:** Replace manual `aws rds promote-read-replica` with Lambda function triggered by RDS event subscription or custom SNS alarm. Saves ~45 seconds.
2. **Lower health-check thresholds:** Change to interval=3s, failure_threshold=2 (6s total detection time vs. 15s).
3. **Reduce DNS TTL:** Set to 30s for faster client-side expiry; pair with app-level connection retry logic (timeout=5s).
4. **Connection pooling strategy:** Apps should pre-warm read-replica endpoint connection pools to standby before switchover, eliminating warm-up time.
5. **RDS parameter groups:** Pre-create identical parameter group on standby; reduces promotion time by ~20s.
6. **Write-ahead logging + S3 backup:** Supplement RDS replication with continuous WAL exports to S3; RPO becomes sub-second if read-replica replication fails.
7. **Chaos engineering:** Run monthly FIS experiments with variable failure modes (network partition vs. instance crash vs. AZ outage) to catch edge cases.

Resume line

> Designed and deployed a multi-region AWS disaster recovery architecture (Terraform, RDS cross-region replication, Route 53 failover routing); validated via live failover simulation achieving **135s RTO** and **1s RPO**, with production roadmap to sub-60s RTO via Lambda-driven promotion and aggressive health-check tuning.